PROPERTIES OF BROWNIAN MOTION

- CONSIDER A "STARTING POINT" OR $t \in [t_0, T]$

$$W_{t_0} = W_0 = 0$$

FORMALLY $\longrightarrow$ $\mathbb{P}[W_{t_0} = 0] = 1$

- FURTHERLY W_t IS ALMOST SURELY CONTINUOUS OR

$$\mathbb{P}[\lim_{n \to \infty} X_n = X] = 1$$

- LASTLY W_t HAS INDIPENDENT AND STATIONARY INCREMENTS

$$W_{t+1} - W_t \sim N(0, (t+1-t))$$

or

$$W_t - W_s \sim N(0, t-s)$$